

Estimation of Quantum Fisher Information by variational quantum circuits on a quantum computer



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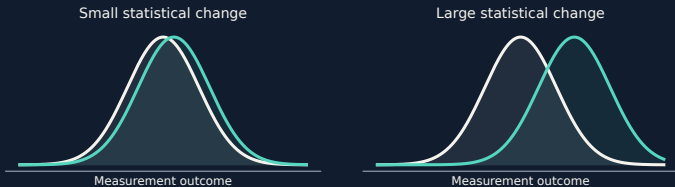
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Fisher information and parameter estimation

Local sensitivity quantifies the distinguishability of nearby parameter values.



Fisher information

How sensitive the observed statistics are to a small parameter change.

Quantum Fisher information (QFI) is the best local information available from the state, optimized over measurements.

Illustrative distributions. QFI is a local benchmark; attainable precision also depends on the measurement and estimation assumptions.

VQSE and principal-component extraction

Variational Quantum State Eigensolver: dominant spectral components of a mixed state.

PCA / COVARIANCE MATRIX

$$C = \sum_i s_i \mathbf{v}_i \mathbf{v}_i^\top$$

Dominant directions capture the largest share of variance.

VQSE / DENSITY MATRIX

$$\rho = \sum_i \lambda_i |v_i\rangle \langle v_i|$$

Dominant state components carry the largest probability weights.

$$V \rho V^\dagger \approx \text{diag}(\lambda_1, \lambda_2, \dots)$$

Train a basis rotation V .
Frequencies estimate eigenvalues;
 $V^\dagger|i\rangle$ recovers eigenvectors.

The retained eigenspace provides a compact input to fidelity-based QFI bounds.

Implemented and benchmarked established VQSE / VQFIE methods: Cerezo et al. (2022); Beckey et al. (2022). No speedup claim.

NUMERICAL MODEL

Local metrology in the transverse-field Ising model

Six interacting spins encode a magnetic field; measurements access a four-spin subsystem.



Periodic spin chain

4 retained / 2 discarded

Ground state Lowest-energy preparation

Depolarized Fixed depolarizing noise

Thermal Field-dependent populations

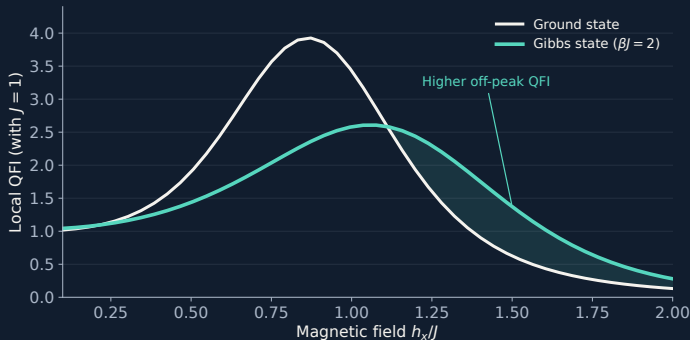
Compare preparation families over the operating field range.

Transverse-field Ising model (TFIM), $N = 6$, $n = 4$, periodic boundaries, $J = 1$. Thesis: NoiseAnalysis, Project 9.

NUMERICAL RESULTS

Thermal preparation and the operating field range

At $\beta J = 2$, the reduced Gibbs state trades peak sensitivity for higher off-peak QFI.



Lower peak

2.61 versus 3.93
for the ground state

Off-peak gain

Useful when the field
is not known to lie
near the optimum.

Preparation should reflect the expected field range, not only the maximum QFI.

Periodic $N = 6$, $n = 4$, $\beta J = 2$. Exact SLD QFI from the thesis model; displayed range $h_x/J \in [0.1, 2]$.

Spatial-mode measurements for exoplanet imaging

A faint companion must be resolved in the presence of a bright reference source.

Direct imaging



Record where each photon lands.

Spatial-mode sorting



Record which spatial mode is occupied.

SPADE changes the measurement basis to reveal weak spatial signals.

Schematic, not experimental imagery. SPADE: spatial-mode demultiplexing. Tsang et al. (2016); Santamaria et al. (2024).

EXPERIMENTAL DATA

Experimental observations and identifiability

The completed ASI Matera experiment supplied aggregate first-order counts.

52,500

count observations across 525 settings
100 repetitions per setting, 10 ms each

Estimation of the mean response
and predictive uncertainty.

Missing observables

Separate modal outputs

Measurements in other bases

Incident flux and failure outcomes

The record supports prediction, but cannot reconstruct the quantum state.

Thesis: ExoplanetExperiment, dataset description. No programmable basis rotations were recorded; experimental TQFI was not obtained.

Classical probabilities and quantum states

A probability vector has a diagonal quantum representation in a chosen basis.

CLASSICAL PROBABILITY EMBEDDING

$$\mathbf{p} = \begin{pmatrix} p_0 \\ p_1 \end{pmatrix} \longmapsto \rho_{\text{diag}} = \begin{pmatrix} p_0 & 0 \\ 0 & p_1 \end{pmatrix}$$

Classical channel: $\mathbf{p}' = R\mathbf{p}$

Redistributes probabilities between outcomes.

QUANTUM COHERENCE

$$\rho = \begin{pmatrix} p_0 & c \\ c^* & p_1 \end{pmatrix}$$

Quantum channel: $\rho' = \mathcal{E}(\rho)$

Acts on populations and coherences.

The embedding preserves measured probabilities; coherences require further data.

Two-outcome illustration: $p_i \geq 0$, $p_0 + p_1 = 1$, $|c|^2 \leq p_0 p_1$. A state vector of probabilities does not determine a quantum channel.

A sequential quantum-to-classical model

Represent propagation, measurement and readout as distinct physical and statistical stages.



Quantum states
populations *and* coherences

Classical probabilities
calibration and count statistics

Coherence and loss / modular calibration / information at each stage

The model connects quantum information limits to observable count statistics.

Thesis: sequential quantum-to-classical framework. Optical loss and inaccessible modes retain explicit failure outcomes in the model.

STATISTICAL RESULTS

Predictive validation of the calibration model

A Gaussian-process correction captures smooth setting-dependent response.

$$\underbrace{\mu(d_a, \epsilon)}_{\text{setting mean}} = \underbrace{\mu_0(d_a, \epsilon)}_{\text{physical response}} \underbrace{\exp[f(d_a, \epsilon) + \xi]}_{\text{calibration correction}}$$

Checkerboard cells

93.5%

95% interval coverage

Distance rows

94.7%

95% interval coverage

Source-ratio columns

93.7%

95% interval coverage

Grouped five-fold validation; all model parameters refitted within each training fold.

Validated target: the 100-repeat setting mean.

Thesis: ExoplanetExperimentAppendix, hierarchical calibration table. Rounded coverage; standardized residual SD 1.00–1.04.

Information retention under restricted readout

Compare classical Fisher information with the source-state QFI on the same setting grid.

Resolved low-mode outcomes

99.68–100%

of source-state QFI retained
as classical Fisher information

Binary readout + cross-talk

0.13–57.0%

under an assumed symmetric
0.35% port-confusion model

Information retention depends on modal resolution and readout fidelity.

Same nine nonzero settings; per photon surviving common loss. Ranges across settings, not confidence intervals. Conditional model.

CONCLUSIONS

Conclusions and experimental outlook

Variational estimation, state preparation and experimentally accessible information.

01 Variational spectral estimation

Implementation and benchmarking of mixed-state QFI bounds.

02 Preparation-dependent sensitivity

Thermal states provide higher off-peak QFI in the studied field range.

03 Experimental modelling and validation

Predictive calibration with explicit limits on quantum-state inference.

Outlook: resolved outputs, calibrated channels and a trainable optical basis.

Fidelity-based bounds and local QFI

The finite-displacement diagnostic and the local QFI are distinct.

$$\mathcal{I}_{\Delta} = \frac{8[1 - F(\rho_{\theta}, \rho_{\theta+\Delta})]}{\Delta^2}$$

Root fidelity F compares nearby states.

Dominant-eigenspace truncation bounds this fidelity quantity.

At regular points, $\mathcal{I}_{\Delta} \rightarrow \mathcal{F}_Q$ as $\Delta \rightarrow 0$.

Finite-step bounds are not automatically pointwise bounds on exact SLD QFI.

Methods: finite-displacement TQFI / SSQFI bounds. The fidelity limit requires regularity at the operating point.

Thermal preparation and data processing

Different preparation families can have different field responses.

$$\rho_{\beta}(h) = \frac{e^{-\beta H(h)}}{\text{Tr } e^{-\beta H(h)}}$$

The field changes eigenvectors *and* thermal populations.

$$\mathcal{F}_Q[\text{Tr}_E \rho_{\beta}(h)] \leq \mathcal{F}_Q[\rho_{\beta}(h)]$$

A thermal preparation is not a fixed channel applied to the ground state.

The advantage compares preparations; it does not reverse a fixed noise channel.

Thesis: NoiseAnalysis, Project 9. Exact finite-system SLD benchmark, not a universal temperature advantage.

Generalization across the setting grid

Target: the mean of 100 repeated counts, in counts per 10 ms.

Split	RMSE	Standardized SD	95% coverage
Checkerboard	0.606	1.004	93.52%
Distance rows	0.734	1.041	94.67%
Source-ratio columns	0.609	1.009	93.71%

Matérn-3/2 GP: smooth calibration + independent setting scatter.
 Individual counts: median Fano factor $\simeq 2.50$ (overdispersed).

Coverage is evidence about predictive uncertainty for means, not channel identification.

Thesis: ExoplanetExperimentAppendix, hierarchical calibration validation and raw-count diagnostics.

Experimental requirements for VQ-SPADE

A proposed optical realization of variational state eigensolving.

- 1 A programmable coherent transformation of spatial modes.
- 2 Resolved modal counts, normalization and failure outcomes.
- 3 Additional overlap measurements for nearby states.
- 4 Independent flux, background and channel calibration.

A future acquisition would turn the learning loop into an optical protocol.

Thesis: proposed VQ-SPADE scheme. Not implemented or tested in the completed experiment.